

KRISTIINANKAUPUNKI FYR, Finland

WMO: 027520

Lat: 62.2036N

Lon: 21.1697E

Elev: 22

StdP: 101.06

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.4	(c) -13.5	(d) -18.6	(e) 0.7	(f) -15.8	(g) -15.9	(h) 0.9	(i) -12.6	(j) 19.6	(k) -0.4	(l) 18.1	(m) -1.0	(n) 4.6	(o) 50	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 3.7	(c) 23.0	(d) 18.7	(e) 21.7	(f) 18.0	(g) 20.1	(h) 17.0	(i) 20.0	(j) 22.4	(k) 18.6	(l) 20.6	(m) 17.6	(n) 19.6	(o) 5.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.8	21.5	17.8	12.8	20.1	16.7	11.9	18.9	57.4	22.4	52.9	20.7	49.5	19.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.4	(b) 14.7	(c) 13.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 16.4	(b) 14.7	(c) 13.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-3.6	-3.3	-1.3	2.4	8.3	12.0	16.3	16.1	12.8	6.7	2.9	-0.2	(6)
(7)		DBStd	7.86	4.60	5.16	3.23	2.36	4.04	3.15	3.09	2.36	2.44	2.96	3.85	4.68	(7)
(8)		HDD10.0	2108	421	372	351	228	83	12	0	1	8	107	212	315	(8)
(9)		HDD18.3	4594	679	605	609	477	311	192	81	77	167	361	462	573	(9)
(10)		CDD10.0	586	0	0	0	1	31	72	196	190	91	4	0	0	(10)
(11)		CDD18.3	29	0	0	0	0	1	3	18	8	0	0	0	0	(11)
(12)		CDH23.3	44	0	0	0	0	1	2	26	16	0	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	0	2	2	0	0	0	0	(13)
(14)	Wind	WSAvg	6.9	7.2	7.3	6.2	6.3	6.5	6.2	5.7	6.0	6.7	7.7	8.2	8.6	(14)
(15)	Precipitation	PrecAvg	620	46	34	32	29	41	56	73	73	57	65	58	57	(15)
(16)		PrecMax	795	98	70	55	60	88	114	135	158	139	154	128	106	(16)
(17)		PrecMin	460	8	2	6	2	6	15	11	16	21	9	10	16	(17)
(18)		PrecStd	82	21	19	13	15	21	22	30	33	33	31	25	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.2	6.0	14.6	22.4	23.3	26.1	25.7	18.7	13.8	10.4	7.6	(19)
(20)			MCWB	3.2	2.6	3.0	8.2	13.8	16.3	19.4	19.1	15.6	11.8	9.0	6.4	(20)
(21)		2%	DB	3.7	3.5	4.1	8.2	19.2	20.8	23.5	22.2	17.0	12.2	9.2	6.5	(21)
(22)			MCWB	2.4	2.1	2.5	5.3	13.9	15.3	19.1	18.9	14.9	10.8	8.5	5.3	(22)
(23)		5%	DB	2.7	2.7	3.2	6.1	16.6	18.4	22.2	20.6	16.4	11.4	8.1	5.4	(23)
(24)			MCWB	1.8	1.7	2.0	3.8	12.7	14.7	18.9	17.7	14.5	9.9	7.1	4.2	(24)
(25)		10%	DB	1.9	2.0	2.5	5.0	13.9	16.3	20.9	19.0	15.7	10.5	7.2	4.6	(25)
(26)			MCWB	1.1	1.3	1.5	3.3	10.5	14.0	18.0	16.7	14.0	9.1	6.3	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.2	4.0	8.5	16.0	17.7	21.4	21.8	16.2	12.2	9.2	6.8	(27)
(28)			MCDB	4.5	3.9	5.0	13.8	18.7	20.2	23.7	23.8	17.7	13.2	10.0	7.3	(28)
(29)		2%	WB	2.7	2.5	2.9	5.7	14.4	16.2	20.5	19.2	15.6	11.2	8.5	5.4	(29)
(30)			MCDB	3.5	3.3	3.6	7.6	18.6	19.2	22.9	21.4	16.7	12.1	9.1	6.3	(30)
(31)		5%	WB	2.0	2.0	2.2	4.5	12.7	15.2	19.4	18.1	15.0	10.3	7.4	4.3	(31)
(32)			MCDB	2.8	2.7	3.0	5.8	16.5	18.1	21.6	20.1	16.1	11.2	7.9	5.2	(32)
(33)		10%	WB	1.3	1.4	1.7	3.6	10.8	13.9	18.2	17.2	14.4	9.3	6.4	3.7	(33)
(34)			MCDB	1.9	2.1	2.5	4.6	13.4	16.2	20.3	18.9	15.5	10.3	7.0	4.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	3.7	3.4	4.4	3.8	3.7	3.1	2.9	2.9	2.8	3.0	(35)
(36)			MCDBR	3.3	3.6	3.5	5.5	7.9	7.3	6.0	4.4	3.5	2.6	2.7	2.9	(36)
(37)			MCWBR	3.0	3.1	2.8	3.4	3.8	3.9	3.4	3.0	2.7	2.8	2.8	2.9	(37)
(38)		5% WB	MCDBR	3.3	3.3	3.3	5.2	7.3	6.3	4.7	3.4	3.0	2.6	2.8	2.8	(38)
(39)			MCWBR	3.1	2.8	2.9	3.5	3.8	4.0	3.3	2.9	2.6	2.9	2.9	2.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.304	0.322	0.361	0.359	0.353	0.377	0.364	0.340	0.323	0.273	0.349	(40)	
(41)		taud	2.119	2.239	2.320	2.338	2.432	2.485	2.442	2.467	2.515	2.392	2.166	2.538	(41)	
(42)		Ebn at Noon	419	648	787	825	860	873	837	818	772	629	387	254	(42)	
(43)		Edh at Noon	34	63	83	101	100	96	98	88	70	55	32	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.17	0.72	2.23	3.85	5.47	6.06	5.54	4.28	2.54	1.01	0.25	0.10	(44)	
(45)		RadStd	0.02	0.14	0.26	0.41	0.49	0.37	0.46	0.36	0.24	0.18	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (43 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-121	N/A	N/A	

Nomenclature: See separate page